

**I TRAINED A NEURAL NETWORK ON 214,425 BANK-
QUARTER EXAMPLES.**

to forecast next-quarter deposit growth.

AND THE MAE WINNER WAS...

PREDICT 0%.

for every bank.

BANK A → 0%

BANK B → 0%

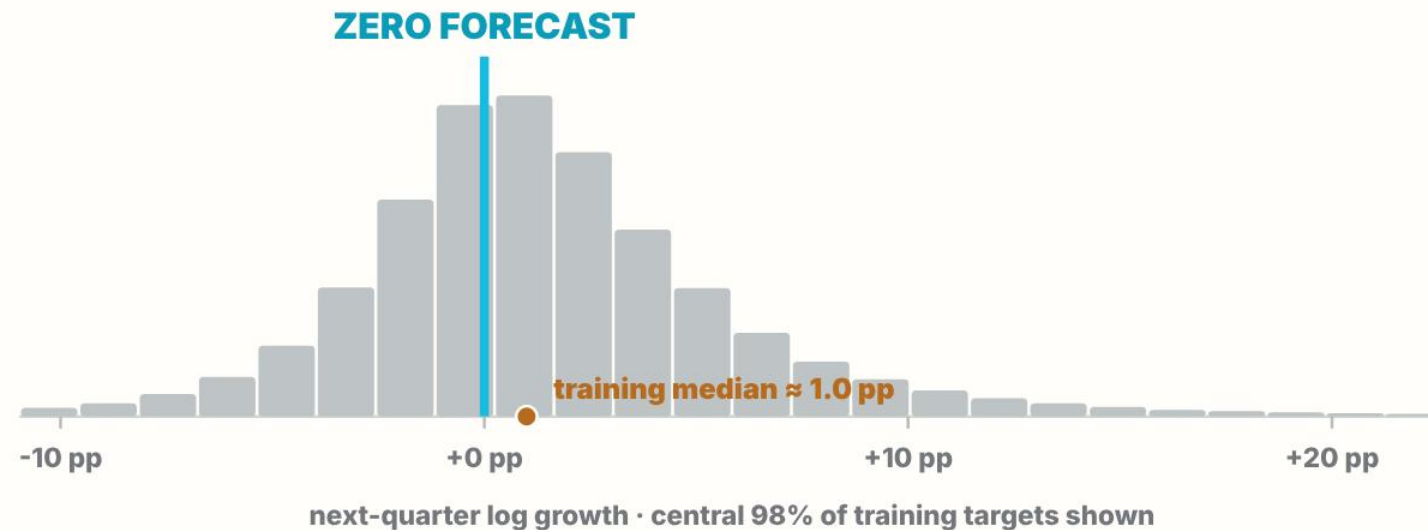
BANK C → 0%

Zero growth: 3.601 pp MAE MLP: 3.627 Ridge: 3.641

MAE = average absolute size of the forecast miss. Lower is better.

WAIT. HOW CAN "0%" WIN ANYTHING?

Because the training target is packed close enough to zero that a constant 0% forecast can rack up a lot of relatively small misses.



So zero growth is lazy. It is **not random guessing.**

Central 98% shown here for readability. The tails stayed in the model data.

SO... WHICH MODEL IS ACTUALLY "BETTER"?

AVERAGE SIZE OF THE MISS

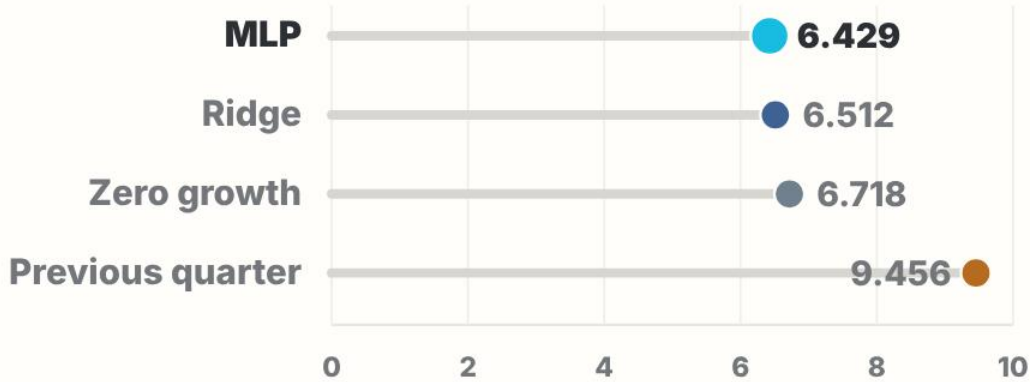
MAE ↓

BIG MISSES HURT MORE

RMSE ↓



Axis starts at 0. Zero, MLP, and Ridge are only 0.04 pp apart.



Axis starts at 0. The MLP has the lowest RMSE.

SAME MODELS. DIFFERENT WINNER.

Okay. So "better" depends on what kind of mistake I care about.

BUT NOW THERE'S A DIFFERENT PROBLEM: EVERY BANK GETS 0%.

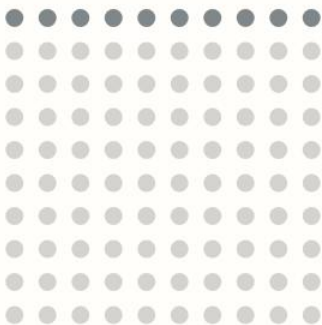


**I CAN REVIEW ONE.
WHICH ONE?**

I can't rank them. They're all tied.

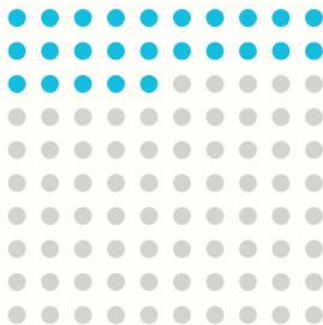
SAME 454 REVIEW SLOTS. VERY DIFFERENT HIT RATES.

RANDOM SELECTION



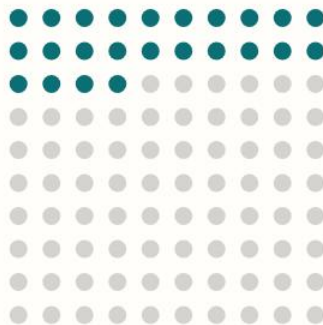
45.4 expected
10.0%

RIDGE



112 hits
24.7%

NEURAL NETWORK



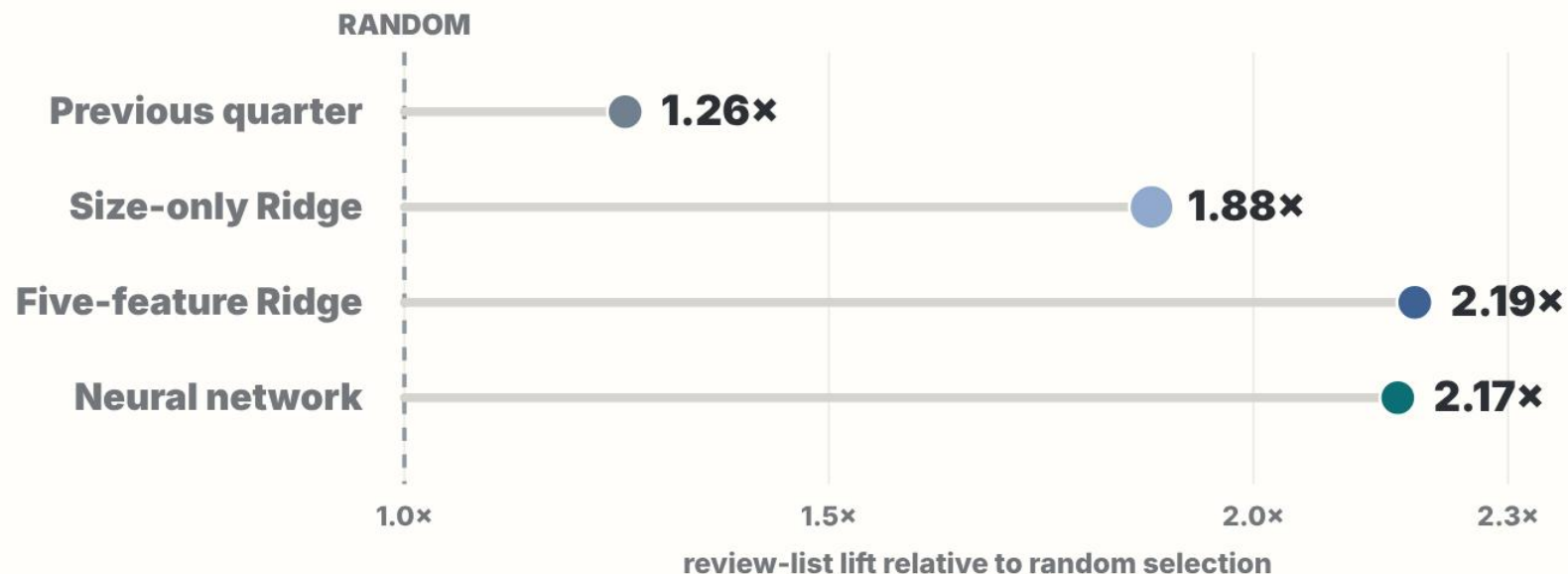
109 hits
24.0%

Hit = a selected bank that later landed in the bottom 10% of next-quarter deposit growth.
Each 100-dot grid shows the share, not individual banks.

NOW THE FORECAST IS DOING SOMETHING ZERO GROWTH PHYSICALLY CAN'T.

SO I TESTED BANK SIZE BY ITSELF.

How far can one obvious feature get on the same review-list task?



SIZE-ONLY RIDGE
1.88x RANDOM

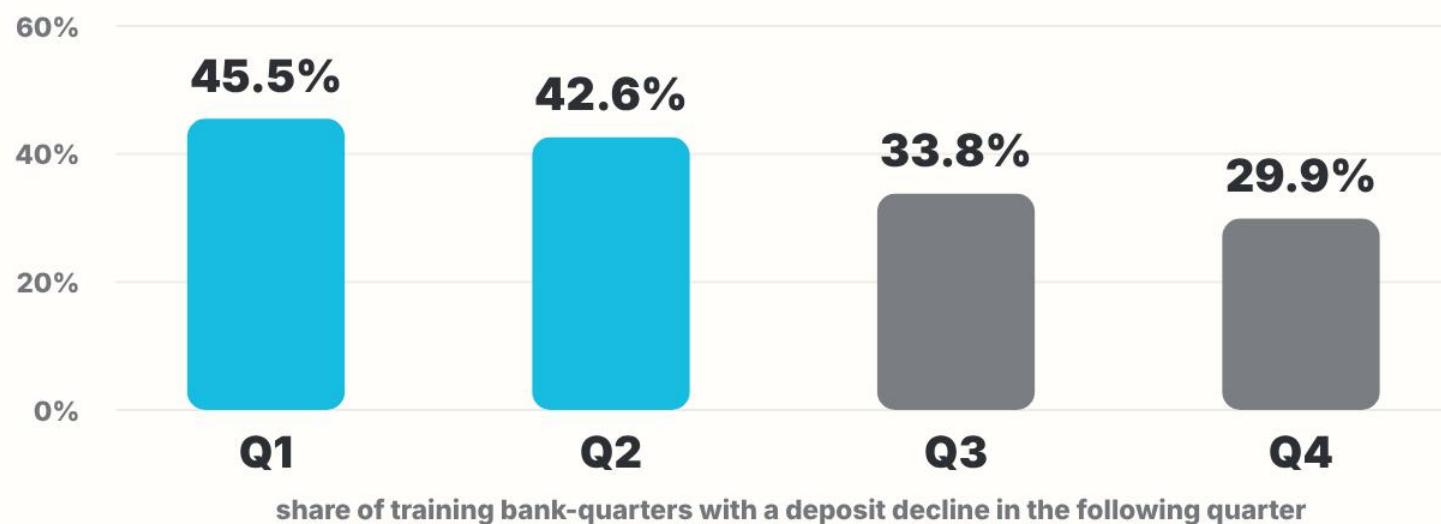
FIVE-FEATURE RIDGE
2.19x RANDOM

This does **not** tell me what share of the full-model result is “caused by size.” It tells me size alone got surprisingly far in these three reused historical quarters.

Size-only Ridge was a follow-up benchmark developed after 2024 had already been examined.

AND THEN THE CALENDAR JOINED THE FIGHT.

In the training years, I grouped predictor quarters and asked: how often did deposits fall next quarter?



SEASONAL MEDIAN
3.500 pp
MAE

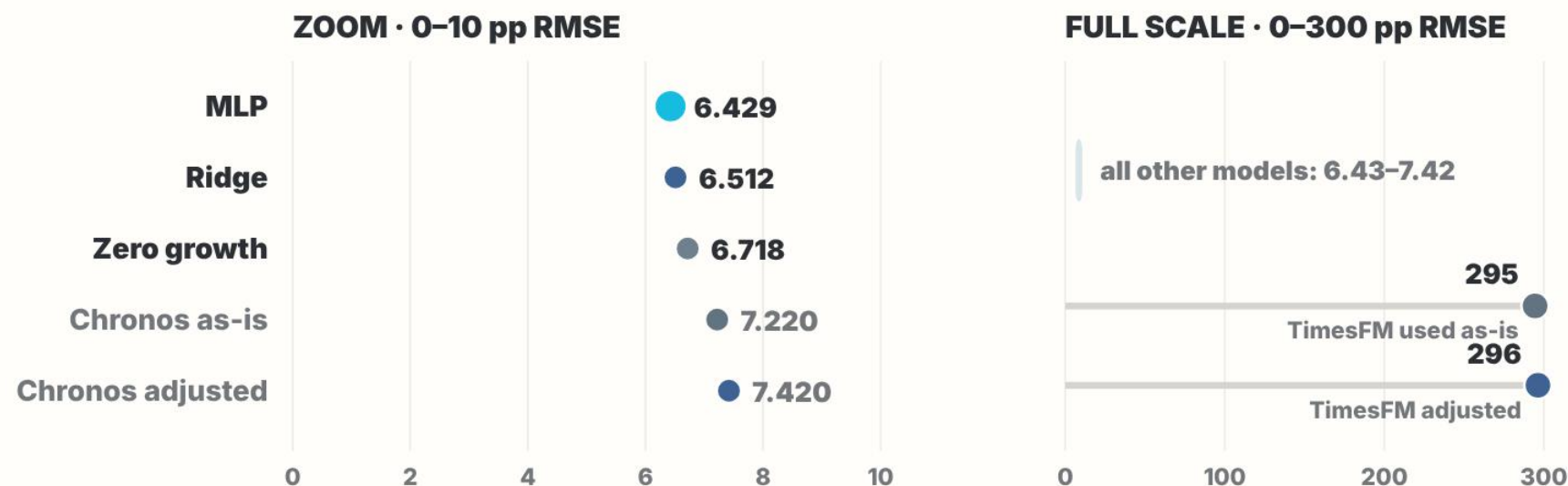
ALWAYS ZERO
3.601 pp
MAE

AND... THE SEASONAL MEDIAN CAME OUT LOWER.

Caveat: I came up with this after already examining 2024. Useful follow-up, not fresh confirmation.

FINE. BRING IN THE PRETRAINED MODELS.

Surely one of THESE beats “predict zero,” right?



...NOPE.

COMPLEXITY DOESN'T GET CREDIT FOR SHOWING UP.

Zero-shot = used as-is. Adjusted = only the forecasting head was adapted on earlier bank data.

Built from my `.ipynb` with **wm-notecards**.